

IILM UNIVERSITY, GREATER NOIDA		PHYSICS LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: Determination of beam divergence of laser		
EXPERIMENT NO. 10	DEPTT: Physics	ISSUE DATE: 12 August, 2026
LABORATORY: 26SOSP101 Semiconductor and Quantum Physics Lab	SEMESTER: I, II	TOTAL PAGE: 3

AIM: Determination of beam divergence of laser

APPARATUS: He-Ne laser optical bench, screen and ruler.

THEORY:

The laser (light amplification by stimulated emission of radiation) is a device that produces a strong beam of coherent photons by stimulated emission. A laser beam is coherent, very narrow and intense. The directionality of the laser beam is expressed in term of the full angle beam divergence, which is twice the angle that the outer edge of the beam makes with the center of the beam as shown in Fig. (1), the divergence tells us how rapidly the beam separates when it is emitted from the laser. Although the divergence angle can be measure in fractions of degrees or radian, the relation between degree and rad is $2\pi = 360\text{degree}$ and $1\text{rad} = 57.3\text{ degree}$, $1\text{mrad} = 0.057\text{ degree}$.

Consider a monochromatic beam of light of “infinite” extent, which passes through a circular aperture of diameter D . The beam will now diverge by an amount dependent on the size of D .

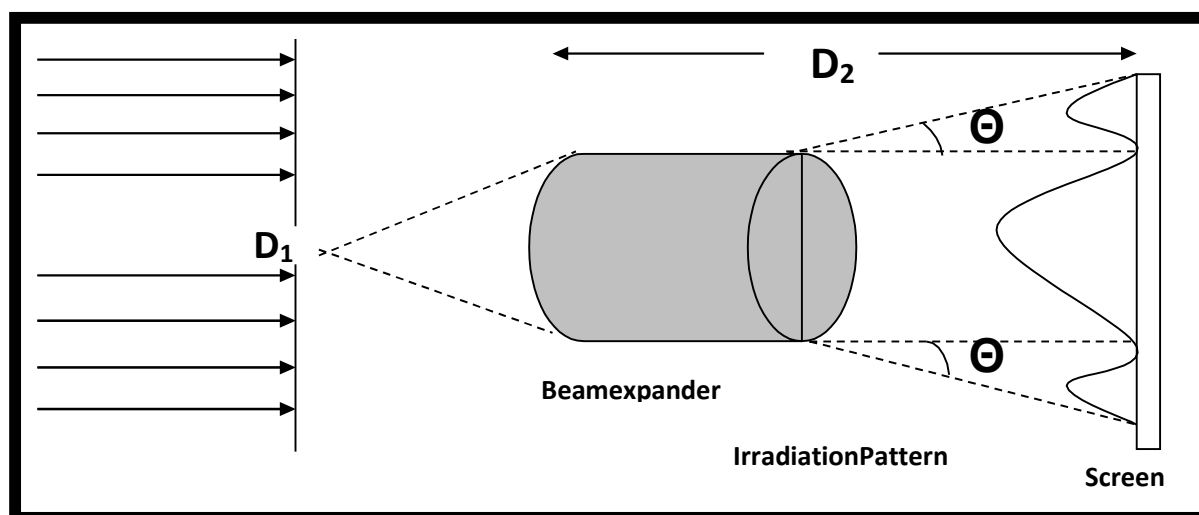
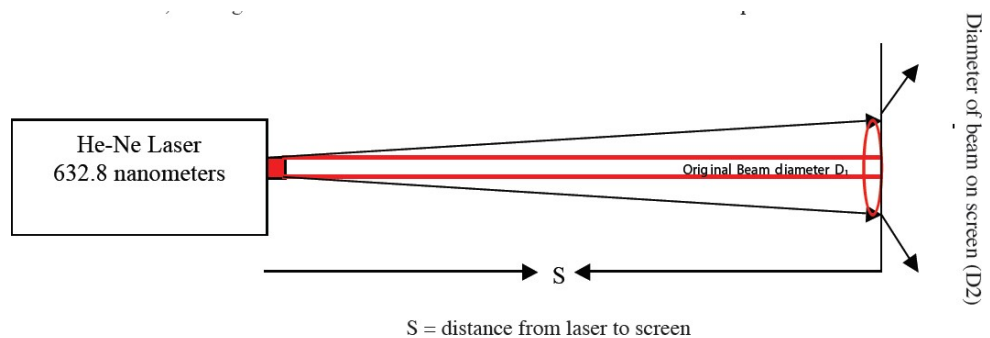


Fig (1): Beam divergence from a circular aperture

PROCEDURE:- Part A :-Without beam expander:

1. Determined (D1) which represented the diameter aperture of He-Ne laser.
2. Place the He-Ne laser at distance of about (S=500 mm) from the screen as in Fig. (2)
3. Determined the diameter of beam on screen (D2).



4. Increase the distance (S) from (500 to 3000) mm and arrange your results as in table below:

S(mm)	D1(mm)	D2(mm)	D3=D2-D1
500			
To			
3000			

1. Plot a graph between D3 as a function of S to find the slope.
2. Slope = $\tan \theta$ Where:
 λ : Wavelength of He-Ne laser=632.8nm
 θ : Divergence angle.

Part B:- With beam expander:

Repeat all the steps in Part A but S are the distance between beam expander and the screen See fig. (1).

RESULT:

Angle of divergence of the beam using He-Ne laser =

OUTCOME:

To understand the importance of laser beam compared to ordinary light

To determine the angle of divergence of the laser beam

PRECAUTION:

1. Do not look directly into the laser beam.
2. Ensure there are no obstacles (dust, fingers, reflections, etc.) in the path of the laser beam to avoid distortion.
3. Perform the experiment in a dim or dark room to clearly observe and measure the laser spot.
4. Switch off the laser when not taking readings to conserve power and prevent accidental exposure.

VIVA QUESTIONS:

1. What is beam divergence?
2. What would happen if you used a flashlight instead of a laser?
3. What is the unit of beam divergence?
4. What is the significance of low divergence in laser applications like fiber optics or laser cutting?